JAI SHARMA

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EDUCATION & COURSEWORK

UC Berkeley, ***B.S. Electrical Engineering & Computer Science (EECS), B.S. Business (Haas)*** **Expected 05/2027**

- M.E.T. Dual Degree Program (<1% acceptance), Regents' & Chancellor's (R&C) Scholarship (top 1% of UC Berkeley), Tech GPA: 3.91
- UC Berkeley EECS Honors program; UC Berkeley Leadership Award (<2% acceptance); Eta Kappa Nu (EECS Honor Society)
- Relevant Coursework: Reinforcement Learning, Information Theory, Advanced Probability, Nonlinear & Discrete Optimization, Control Theory, Linear Programming & Network Flows, Operating Systems, Computer Architecture, Financial Engineering
- Clubs: Applied Statistics (ML projects for Walmart, Spotify, NOAA); Space Enterprise at Berkeley (rocket thrust curve optimization)

INDUSTRY EXPERIENCE

Databricks Inc, ***PhD-level Data Science Intern*** **05/2026 to Present**

- Built agentic workflows to automate quantitative finance analysis; improved modeling of holiday effects by 20% for forecasts

Palo Alto Networks, ***PhD-level AI Research Intern*** **05/2025 to 08/2025**

- Researched LLM red teaming strategies for AI safety; boosted prompt injection detection model coverage by 50% on new attacks
- Patented LLM jailbreak protection by training adversarial models; uses gradient-suffix, Reinforcement Learning, & agentic attacks

BioMiner AI (NVIDIA Inception startup), ***Chief AI Officer*** **09/2024 to 04/2025**

- Co-founder of startup developing foundational models to embed multimodal patient data and identify therapeutic biomarkers
- 1st author on patent; clients include Novo Nordisk, St. Jude Children's Hospital, BioSkryb Genomics, UFlorida, J&J Labs, etc.

IBM Research, ***Machine Learning Researcher for Meta Learning*** **05/2024 to 10/2024**

- Investigated LLM hyperparameter transferability across 1000+ language tasks using Bayesian optimization with statistical prior
- Performed LoRA fine-tuning experiments on 7B LLMs using Parameter Efficient Fine Tuning (PEFT), QLoRA, TensorBoard

Anaflash Inc (Berkeley Skydeck startup, backed by NASA), ***Machine Learning Engineer for TinyML*** **01/2024 to 06/2024**

- Researched 4-bit quantization to detect anomalous farm animal coughs for edge AI chip with <10 mW power consumption
- Secured 1st place Best Prototype Award, TinyML 2024 Research Conference (for NAND flash based neural network accelerator)

RESEARCH EXPERIENCE

Berkeley AI Research (BAIR), Prof. Dawn Song Group, ***Machine Learning Researcher*** **09/2025 to Present**

- Identified vulnerabilities in diffusion language models through reverse generating prompts to elicit harmful target responses

Berkeley AI Research (BAIR), Prof. Ion Stoica Group, ***Machine Learning Researcher*** **03/2024 to 01/2025**

- Developed Reinforcement Learning methods to automate 100s of web navigation tasks on Reddit, e-commerce websites, etc
- Applied actor-critic models to boost zero-shot performance of multimodal LLMs on 70+ GB sized Visual WebArena dataset

Berkeley AI Research (BAIR), Prof. Masayoshi Tomizuka Group, ***Machine Learning Researcher*** **03/2024 to 12/2024**

- Created framework to perform local feature transfer between images using 70M parameter U-Net for stable diffusion

Stanford University, Dr. William Robinson Lab, ***Bioinformatics Engineer*** **06/2021 to 06/2023**

- Conducted biostatistical analysis of mRNA seq data with 40000+ genes & 1000+ samples for ScienceDirect publication

PUBLICATIONS & TEACHING

Selected Research Publications

- Diffusion Investigator Agents **(2026)**; Neural Estimation of Pairwise Mutual Information in Masked Diffusion Models **(2025)**
- Autonomous Web Navigation using Multimodal Agents **(2024)**; Deep Learning for Geocentric Pose Prediction **(IEEE PRAI 2022)**
- RNA-seq Characterization of Histamine-Releasing Mast Cells as Potential Therapeutic Target of Osteoarthritis **(ScienceDirect 2022)**
- Explainable ML diagnosis models for CVD **(IEEE iBioMed 2022)**; Dropout Regularization vs Model Complexity in NNs **(arXiv 2021)**

Patents	Adversarial Reinforcement of an LLM Firewall	**09/2025**
	User-context Guided Agentic Artificial Intelligence System for Binary File Reverse Engineering	**08/2025**
	Systems for Multimodal Biological Data Translation and Embedding Representations	**10/2024**
Textbook Publications	Diffusion from First Principles **(In Progress)**	**10/2025**
	Introduction to Deep Learning ***(ISBN 979-8767404476)***	**11/2021**
	Conquering the AMC 8, Math Competition Preparation ***(ISBN 979-8562144157)***	**11/2020**

UC Berkeley, ***Teaching Assistant*** **08/2024 to Present**

- CS 70 (Discrete Math & Probability): TA, Academic Intern, CS Senior Mentor; IEOR 162 (Linear Programming & Network Flows): TA

AWARDS

- **Kleiner Perkins Fellow** (<1% acceptance), **Greylock X finalist** (<1%), presented to **NYSE** about dynamic GPU pricing models
- **PANW Internal R&D Hackathon:** 1st place of 300+ participants, **Cloudflare AI Agents Hackathon:** 2nd place of 400+ participants
- **IMC Trading Prosperity Competition:** top 1% of 20000+ teams, **Citadel Terminal Competition** (<1%), **D. E. Shaw Connect Invitee**
- **PROMYS:** Boston University Mathematics Program (<5% acceptance rate), **AIME:** Top 2.5% math exam score of 300000 students